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Environment	Kali Linux / Ubuntu	Submission	E-Campus

LINUX FIREWALL AND SECURE REMOTE ACCESS REPORT

TASK 1 REPORT

EVIDENCE:

```
dyahayaaddra11@ubuntu: ~/core102_lab4/web
Session Actions Edit View Help
Last login: Tue Sep 29 14:08:40 2026 from 1
dyahayaaddra11@ubuntu:~$ date
Tue Sep 29 02:26:08 PM UTC 2026
dyahayaaddra11@ubuntu:~$ hostname
ubuntu
dyahayaaddra11@ubuntu:~$ ip -br addr
lo UNKNOWN 127.0.0.1/8
ens33 UP 192.168.149
ens37 UP 192.168.113
dyahayaaddra11@ubuntu:~$ ip route
default via 192.168.149.2 dev ens33 proto d
192.168.113.0/24 dev ens37 proto kernel sco
192.168.113.1 dev ens37 proto dhcp scope li
192.168.149.0/24 dev ens33 proto kernel sco
192.168.149.2 dev ens33 proto dhcp scope li
dyahayaaddra11@ubuntu:~$ sudo ss -tulnp
[sudo: authenticate] Password:
Netid State Recv-Q Send-Q Local Ad
Port
Process
udp UNCONN 0 0 127.0
*
users:(("systemd-resolve",pid=623,fd=18))
udp UNCONN 0 0 127.0.0.
*
users:(("systemd-resolve",pid=623,fd=16))
udp UNCONN 0 0 192.168.149.130%
*
users:(("systemd-network",pid=890,fd=37))
udp UNCONN 0 0 192.168.113.129%
*

L-$ curl -v http://192.168.113.129:8080/
* Trying 192.168.113.129:8080...
* Established connection to 192.168.113.129 (192
.168.113.129 port 8080) from 192.168.113.128 por
t 44552
* using HTTP/1.x
> GET / HTTP/1.1
> Host: 192.168.113.129:8080
> User-Agent: curl/8.20.0
> Accept: */*
* Request completely sent off
* HTTP 1.0, assume close after body
< HTTP/1.0 200 OK
< Server: SimpleHTTP/0.6 Python/3.14.4
< Date: Tue, 29 Sep 2026 14:50:34 GMT
< Content-type: text/html
< Content-Length: 33
< Last-Modified: Tue, 29 Sep 2026 14:44:46 GMT
<
CORE102 authorised firewall test
* shutting down connection #0
(kali@kali)-[~]
$ nc -vz 192.168.113.129 8080
192.168.113.129: inverse host lookup failed: Unk
nown host
(UNKNOWN) [192.168.113.129] 8080 (http-alt) open
(kali@kali)-[~]
$

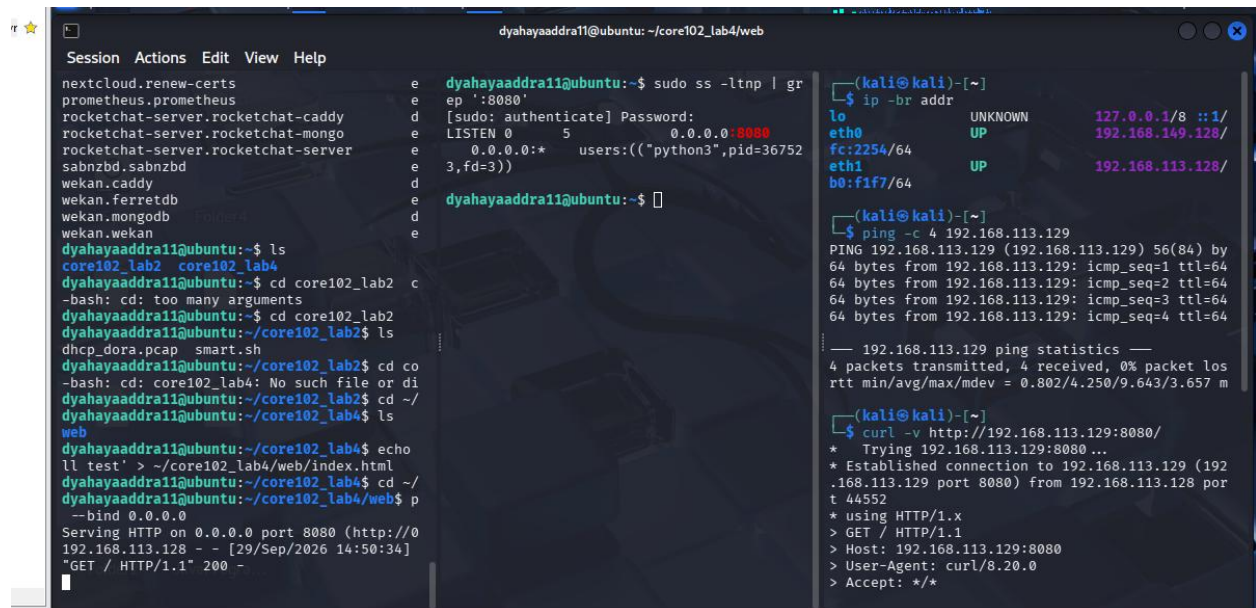
num pkts bytes target prot opt in out source destin
Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
num pkts bytes target prot opt in out source destin
1 0 0 ACCEPT all -- * * 10.1.0.0/16 0.0.0.
0/0 /* generated for MicroK8s pods */
2 0 0 ACCEPT all -- * * 0.0.0.0/0 10.1.0
0/16 /* generated for MicroK8s pods */
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
num pkts bytes target prot opt in out source destin
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num pkts bytes target prot opt in out source destin
Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
num pkts bytes target prot opt in out source destin
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
num pkts bytes target prot opt in out source destin
dyahayaaddra11@ubuntu:~/core102_lab4$
```

RESPONSE:

Port and protocol	Bind address	Process	Expected purpose	Keep or remove
Port and protocol	Bind address	Process	Expected purpose	Keep or remove
22/tcp	0.0.0.0 and [::]	sshd	Remote administration over SSH	Keep. Required for this lab. Restrict to the lab subnet 192.168.113.0/24 with UFW.
80/tcp	(all interfaces)	httpd	Web server	Remove or block. Not needed for this lab, and it is exposed to the network. UFW default-deny will block it.
9090/tcp	(all interfaces)	prometheus	Monitoring metrics and web interface	Remove or block. Not needed for this lab, and it exposes system metrics to the network.

TASK 2 REPORT

EVIDENCE:



```
dyahayaaddra11@ubuntu: ~/core102_lab4/web
Session Actions Edit View Help
nextcloud.renew-certs
prometheus.prometheus
rocketchat-server.rocketchat-caddy
rocketchat-server.rocketchat-mongo
rocketchat-server.rocketchat-server
sabnzbd.sabnzbd
wekan.caddy
wekan.ferretdb
wekan.mongodb
wekan.wekan
dyahayaaddra11@ubuntu:~$ ls
core102_lab2 core102_lab4
dyahayaaddra11@ubuntu:~$ cd core102_lab2
-bash: cd: too many arguments
dyahayaaddra11@ubuntu:~$ cd core102_lab2
dyahayaaddra11@ubuntu:~/core102_lab2$ ls
dhcp_dora.pcap smart.sh
dyahayaaddra11@ubuntu:~/core102_lab2$ cd co
-bash: cd: core102_lab4: No such file or di
dyahayaaddra11@ubuntu:~/core102_lab2$ cd ~/
dyahayaaddra11@ubuntu:~/core102_lab4$ ls
web
dyahayaaddra11@ubuntu:~/core102_lab4$ echo
ll test' > ~/core102_lab4/web/index.html
dyahayaaddra11@ubuntu:~/core102_lab4$ cd ~/
dyahayaaddra11@ubuntu:~/core102_lab4/web$ p
--bind 0.0.0.0
Serving HTTP on 0.0.0.0 port 8080 (http://0
192.168.113.128 - - [29/Sep/2026 14:50:34]
"GET / HTTP/1.1" 200 -

dyahayaaddra11@ubuntu:~$ sudo ss -ltnp | gr
ep ':8080'
[sudo: authenticate] Password:
LISTEN 0 5 0.0.0.0:8080
0.0.0.0:* users:(("python3",pid=36752
3,fd=3))
dyahayaaddra11@ubuntu:~$

(kali@kali)-[~]
$ ip -br addr
lo UNKNOWN 127.0.0.1/8 ::1/
eth0 UP 192.168.149.128/
fc:2254/64
eth1 UP 192.168.113.128/
bo:f1f7/64

(kali@kali)-[~]
$ ping -c 4 192.168.113.129
PING 192.168.113.129 (192.168.113.129) 56(84) by
64 bytes from 192.168.113.129: icmp_seq=1 ttl=64
64 bytes from 192.168.113.129: icmp_seq=2 ttl=64
64 bytes from 192.168.113.129: icmp_seq=3 ttl=64
64 bytes from 192.168.113.129: icmp_seq=4 ttl=64
— 192.168.113.129 ping statistics —
4 packets transmitted, 4 received, 0% packet los
rtt min/avg/max/mdev = 0.802/4.250/9.643/3.657 m

(kali@kali)-[~]
$ curl -v http://192.168.113.129:8080/
* Trying 192.168.113.129:8080 ...
* Established connection to 192.168.113.129 (192
.168.113.129 port 8080) from 192.168.113.128 por
t 44552
* using HTTP/1.x
> GET / HTTP/1.1
> Host: 192.168.113.129:8080
> User-Agent: curl/8.20.0
> Accept: /*/*
```

RESPONSE: In this task, I created a temporary web server on the Ubuntu server using Python and ran it on port 8080. I then checked that the service was running and tested it from my Kali machine using curl and nc. The connection was successful, and I received the expected test message. This confirmed that both the web service and the network connection between the two machines were working properly.

What I learned: I learned how to check if a port is listening and how to test communication between a server and a client using tools such as ss, curl, and nc.

TASK 3 REPORT

EVIDENCE:

```
^C
Keyboard interrupt received, exiting.
dyahayaaddra11@ubuntu:~/core102_lab4/web$ sudo ufw status verbose
[sudo: authenticate] Password:
Status: inactive
dyahayaaddra11@ubuntu:~/core102_lab4/web$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
dyahayaaddra11@ubuntu:~/core102_lab4/web$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
dyahayaaddra11@ubuntu:~/core102_lab4/web$ sudo ufw allow from 192.168.113.0/24 to any port 22 proto tcp comment 'CORE102 SSH'
Rules updated
dyahayaaddra11@ubuntu:~/core102_lab4/web$ sudo ufw allow from 192.168.113.128 to any port 8080 proto tcp comment 'CORE102 test web'
Rules updated
dyahayaaddra11@ubuntu:~/core102_lab4/web$ sudo ufw deny 23/tcp comment 'Block Telnet'
Rules updated
Rules updated (v6)
dyahayaaddra11@ubuntu:~/core102_lab4/web$ sudo ufw show added
Added user rules (see 'ufw status' for running firewall):

dyahayaaddra11@ubuntu:~$ sudo ss -ltnp | grep ':8080'
[sudo: authenticate] Password:
LISTEN 0      5            0.0.0.0:8080
          0.0.0.0:*    users:(("python3",pid=367523,fd=3))

dyahayaaddra11@ubuntu:~$
```

```

ing firewall):
ufw allow from 192.168.113.0/24 to any port
22 proto tcp comment 'CORE102 SSH'
ufw allow from 192.168.113.128 to any port
8080 proto tcp comment 'CORE102 test web'
ufw deny 23/tcp comment 'Block Telnet'
dyahayaaddra11@ubuntu:~/core102_lab4/web$ sudo ufw enable
Command may disrupt existing ssh connection
s. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
dyahayaaddra11@ubuntu:~/core102_lab4/web$ sudo ufw status numbered
Status: active

To
From
--
[ 1] 22/tcp
192.168.113.0/24      ALLOW IN
# CORE102 SSH
[ 2] 8080/tcp
192.168.113.128      ALLOW IN
# CORE102 test
web
[ 3] 23/tcp
Anywhere             DENY IN
# Block Telnet
[ 4] 23/tcp (v6)
Anywhere (v6)        DENY IN
# Block Telnet

dyahayaaddra11@ubuntu:~/core102_lab4/web$

```

```

dyahayaaddra11@ubuntu:~$ sudo ss -ltnp | grep ':8080'
[sudo: authenticate] Password:
LISTEN 0      5            0.0.0.0:*      0.0.0.0:8080
          users:((("python3",pid=36752
3,fd=3))
dyahayaaddra11@ubuntu:~$

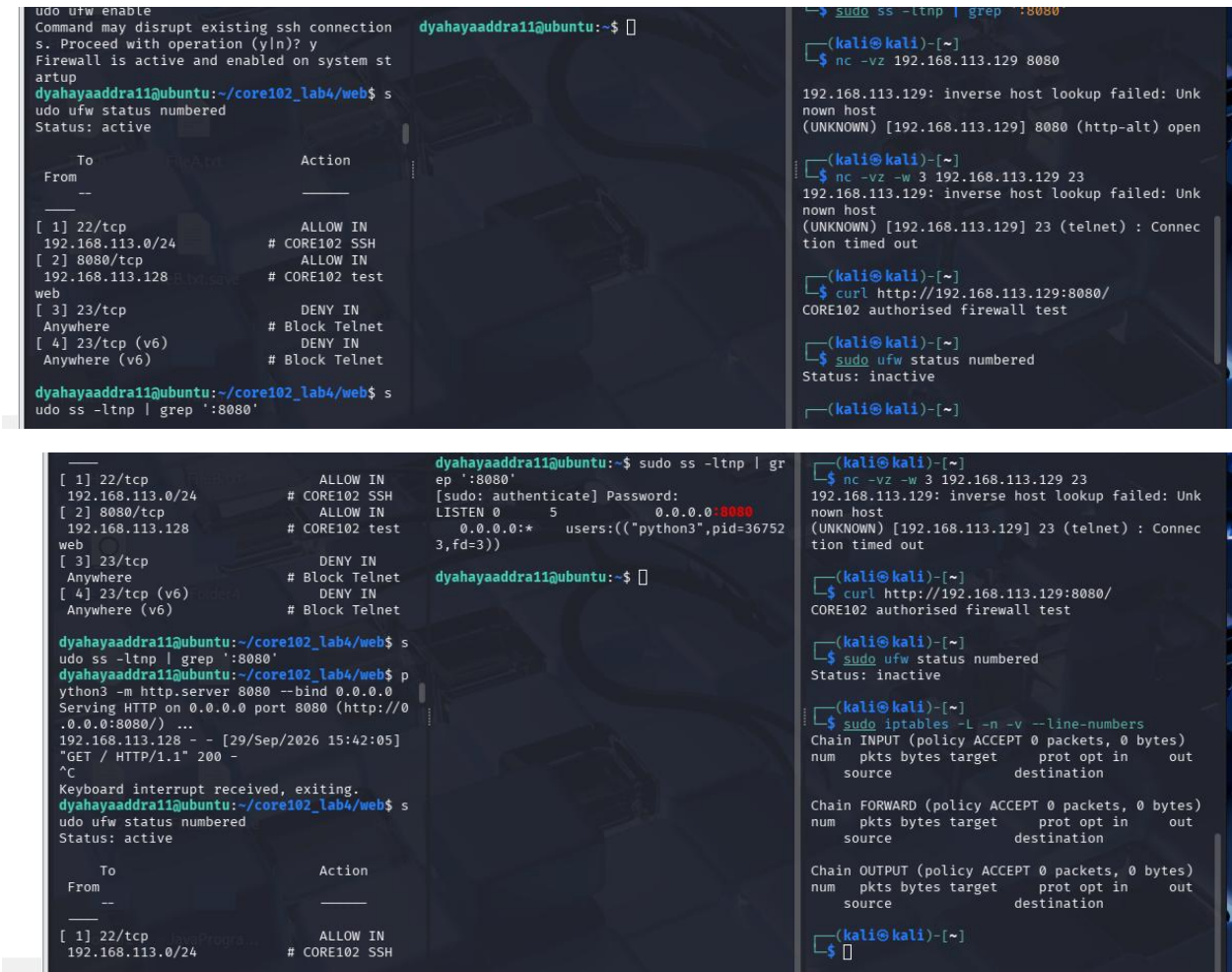
```

RESPONSE: In this task, I configured the UFW firewall on Ubuntu. I set incoming connections to be denied by default while allowing outgoing connections. I then allowed SSH access from my lab network, allowed the temporary web service only from my Kali machine, and blocked Telnet on port 23. After checking the rules, I enabled the firewall and confirmed that the rules were active.

I learned how a firewall controls incoming and outgoing network traffic. I also learned why SSH should be allowed before enabling a default-deny firewall and how to restrict access to a service to only an authorised machine.

TASK 4 REPORT

EVIDENCE:



RESPONSE:

Test	Expected	Observed	Matching rule or explanation
SSH from client	Allowed	Open	UFW rule [1] allows TCP port 22 from 192.168.113.0/24.
HTTP 8080 from authorised client	Allowed	Open	UFW rule allowed TCP port 8080 from Kali 192.168.113.128.
HTTP 8080 from other client	Denied	Not tested	Only 192.168.113.128 was authorised to access port 8080.
Telnet 23	Denied	Connection timed out	UFW rule [2] denies TCP port 23 from anywhere.

In this task, I tested the firewall from the Kali client to make sure the rules configured in Task 3 were working correctly. SSH and the temporary web service on port 8080 were accessible as expected, while the Telnet port was blocked.

I also checked the firewall status and the underlying iptables rules. The packet counters confirmed that the firewall was processing the test traffic.

I also learned that a firewall does not only need to be configured; it must also be tested from the client side to confirm that allowed services work and unwanted traffic is blocked.

TASK 5 REPORT

EVIDENCE:

```
dyahayaaddra11@ubuntu:~/core102_lab4/web$ sudo iptables -N CORE102_LAB
dyahayaaddra11@ubuntu:~/core102_lab4/web$ sudo iptables -A CORE102_LAB -p tcp --dport 23 -j DROP
dyahayaaddra11@ubuntu:~/core102_lab4/web$ sudo iptables -A CORE102_LAB -p tcp --dport 22 -j ACCEPT
dyahayaaddra11@ubuntu:~/core102_lab4/web$ sudo iptables -L CORE102_LAB -n -v --line-numbers
# Warning: iptables-legacy tables present, use iptables-legacy to see them
Chain CORE102_LAB (0 references)
num  pkts bytes target    prot opt in
out   source                destination
1      0      0 DROP      tcp  --  *
*      0.0.0.0/0              0.0.0.0/0
tcp dpt:23
2      0      0 ACCEPT    tcp  --  *
```

```

mbers
# Warning: iptables-legacy tables present,
use iptables-legacy to see them
Chain CORE102_LAB (0 references)
num  pkts bytes target    prot opt in
out      source      destination
1      0      0 DROP      tcp  --  *
*      0.0.0.0/0    0.0.0.0/0
tcp dpt:23
2      0      0 ACCEPT    tcp  --  *
*      0.0.0.0/0    0.0.0.0/0
tcp dpt:22
dyahayaaddra11@ubuntu:~/core102_lab4/web$ s
udo iptables -S CORE102_LAB
# Warning: iptables-legacy tables present,
use iptables-legacy to see them
-N CORE102_LAB
-A CORE102_LAB -p tcp -m tcp --dport 23 -j
DROP
-A CORE102_LAB -p tcp -m tcp --dport 22 -j
ACCEPT
dyahayaaddra11@ubuntu:~/core102_lab4/web$

```

```

out      source      destination
1      0      0 DROP      tcp  --  *
*      0.0.0.0/0    0.0.0.0/0
tcp dpt:23
2      0      0 ACCEPT    tcp  --  *
*      0.0.0.0/0    0.0.0.0/0
tcp dpt:22
dyahayaaddra11@ubuntu:~/core102_lab4/web$ s
udo iptables -S CORE102_LAB
# Warning: iptables-legacy tables present,
use iptables-legacy to see them
-N CORE102_LAB
-A CORE102_LAB -p tcp -m tcp --dport 23 -j
DROP
-A CORE102_LAB -p tcp -m tcp --dport 22 -j
ACCEPT
dyahayaaddra11@ubuntu:~/core102_lab4/web$ sudo iptables -F CORE10
2_LAB
dyahayaaddra11@ubuntu:~/core102_lab4/web$ sudo iptables -X CORE10
2_LAB
dyahayaaddra11@ubuntu:~/core102_lab4/web$ sudo iptables -L CORE10
2_LAB 2>81 || echo "Custom chain removed"
# Warning: iptables-legacy tables present, use iptables-legacy to
see them
iptables: No chain/target/match by that name.
Custom chain removed
dyahayaaddra11@ubuntu:~/core102_lab4/web$

```

RESPONSE:

In this task, I created a temporary iptables chain called CORE102_LAB and added rules to block Telnet on port 23 and allow SSH on port 22. I inspected the rules and confirmed that the chain had zero references, meaning it was not connected to the active firewall and therefore did not affect normal traffic.

After testing the configuration, I flushed the rules and deleted the temporary chain. I then verified that the chain had been completely removed.

I also learned that custom iptables chains can be used to organize firewall rules. A chain does not affect traffic simply because it exists; it must be connected to a main chain such as INPUT, OUTPUT, or FORWARD. I also learned the importance of cleaning up temporary firewall rules after testing.

TASK 6 REPORT

EVIDENCE:

```
2 * 0.0.0.0/0 ACCEPT tcp -- 0.0.0.0/0
tcp dpt:22
dyahayaaddra11@ubuntu:~/core102_lab4/web$ s
udo iptables -S CORE102_LAB
# Warning: iptables-legacy tables present,
use iptables-legacy to see them
-N CORE102_LAB
-A CORE102_LAB -p tcp -m tcp --dport 23 -j
DROP
-A CORE102_LAB -p tcp -m tcp --dport 22 -j
ACCEPT
dyahayaaddra11@ubuntu:~/core102_lab4/web$ sudo iptables -F CORE10
2_LAB
dyahayaaddra11@ubuntu:~/core102_lab4/web$ sudo iptables -X CORE10
2_LAB
dyahayaaddra11@ubuntu:~/core102_lab4/web$ sudo iptables -L CORE10
2_LAB 2>61 || echo "Custom chain removed"
# Warning: iptables-legacy tables present, use iptables-legacy to
see them
iptables: No chain/target/match by that name.
Custom chain removed
dyahayaaddra11@ubuntu:~/core102_lab4/web$ ^C
dyahayaaddra11@ubuntu:~/core102_lab4/web$ mkdir -p ~/.ssh
chmod 700 ~/.ssh
dyahayaaddra11@ubuntu:~/core102_lab4/web$ ssh-copy-id -i ~/.ssh/c
ore102_ed25519.pub dyahayaaddra11@192.168.113.129
/usr/bin/ssh-copy-id: ERROR: failed to open ID file '/home/dyahay
aaddra11/.ssh/core102_ed25519.pub': No such file
dyahayaaddra11@ubuntu:~/core102_lab4/web$

(kali@kali)-[~]
$ mkdir -p ~/.ssh
(kali@kali)-[~]
$ chmod 700 ~/.ssh
(kali@kali)-[~]
$ ssh-keygen -t ed25519 -a 64 -f ~/.ssh/core10
2_ed25519 -C "CORE102 authorised lab"
Generating public/private ed25519 key pair.
Enter passphrase for "/home/kali/.ssh/core102_ed
25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/kali
/.ssh/core102_ed25519
Your public key has been saved in /home/kali/.ss
h/core102_ed25519.pub
The key fingerprint is:
SHA256:ZuMO35Vco8vQyxCSXlJlR7/IGlmHcYdA8bHJLK/
hk CORE102 authorised lab
The key's randomart image is:
+--[ED25519 256]--+
|      +-+*Bo |
|      oo**+= |
|      oo+=+ooo |
|      . = ... + +
|      So.  o *o |
|      o.E = o |
|      o.* . |
```

```
(kali㉿kali)-[~]  
$ ssh-copy-id -i ~/.ssh/core102_ed25519.pub dyahayaaddra11@192.168.113.129  
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/kali/.ssh/core102_ed25519.pub"  
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed  
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys  
Enter passphrase for key '/home/kali/.ssh/id_ed25519':  
Enter passphrase for key '/home/kali/.ssh/id_ed25519':  
Enter passphrase for key '/home/kali/.ssh/id_ed25519':  
dyahayaaddra11@192.168.113.129's password:  
  
Number of key(s) added: 1  
  
Now try logging into the machine, with: "ssh -i /home/kali/.ssh/core102_ed25519 'dyahayaaddra11@192.168.113.129'"  
and check to make sure that only the key(s) you wanted were added.  
  
(kali㉿kali)-[~]
```

```

Last login: Tue Sep 29 14:49:13 2026 from 192.168.113.128
rdyahayaaddra11@ubuntu:~$ sudo journalctl -u ssh --since '10 minutes ago' --no-pager | tail -40
[sudo: authenticate] Password:
Sep 29 16:20:26 ubuntu sshd-session[542669]: Connection closed by authenticating user dyahayaaddra11 192.168.113.128 port 36832 [preauth]
Sep 29 16:20:26 ubuntu sshd-session[542699]: Connection closed by authenticating user dyahayaaddra11 192.168.113.128 port 36836 [preauth]
Sep 29 16:20:57 ubuntu sshd-session[542726]: Accepted password for dyahayaaddra11 from 192.168.113.128 port 36848 ssh2
Sep 29 16:20:57 ubuntu sshd-session[542726]: pam_unix(sshd:session): session opened for user dyahayaaddra11(uid=1000) by dyahayaaddra11(uid=0)
Sep 29 16:22:28 ubuntu sshd-session[544361]: Accepted publickey for dyahayaaddra11 from 192.168.113.128 port 56690 ssh2: ED25519 SHA256:/ZuMO35Vco8vQyxCSXlJlR7/IGGlmHcYdA8bHJLK/hk
Sep 29 16:22:28 ubuntu sshd-session[544361]: pam_unix(sshd:session): session opened for user dyahayaaddra11(uid=1000) by dyahayaaddra11(uid=0)
dyahayaaddra11@ubuntu:~$ stat -c '%a %n' ~/.ssh
~/.ssh/authorized_keys
700 /home/dyahayaaddra11/.ssh
600 /home/dyahayaaddra11/.ssh/authorized_keys
dyahayaaddra11@ubuntu:~$

```

RESPONSE:

In this task, I created an Ed25519 SSH key pair on the Kali client and copied the public key to the Ubuntu server. I then used the private key from Kali to successfully log in to Ubuntu.

I checked the SSH server logs and confirmed that the login was accepted using the ED25519 public key. I also checked the permissions of the SSH directory and authorized_keys file and confirmed they were properly restricted.

I also learned that SSH keys provide a secure way to authenticate without sending the account password during login. The private key must remain secret, while the public key can be placed on the server. I also learned that correct file permissions are important for SSH security.

TASK 7 REPORT

EVIDENCE:

```
dyahayaaddra11@ubuntu: ~  
Session Actions Edit View Help  
* Canonical Workshop gives developers fast, composable, reproducible  
secure developer environments that are perfect for agentc workflo  
https://ubuntu.com/workshop  
Expanded Security Maintenance for Applications is not enabled.  
8 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
*** System restart required ***  
Last login: Tue Sep 29 16:22:33 2026 from 192.168.113.128  
dyahayaaddra11@ubuntu:~$ ls  
core102_lab2 core102_lab4  
dyahayaaddra11@ubuntu:~$ cd core102_lab4  
dyahayaaddra11@ubuntu:~/core102_lab4$ ls  
web  
dyahayaaddra11@ubuntu:~/core102_lab4$ sudo cp /etc/ssh/sshd_config /e  
tc/ssh/sshd_config.core102.bak  
[sudo: authenticate] Password:  
dyahayaaddra11@ubuntu:~/core102_lab4$ sudo nano /etc/ssh/sshd_config.  
d/90-core102-hardening.conf  
dyahayaaddra11@ubuntu:~/core102_lab4$ sudo sshd -t  
dyahayaaddra11@ubuntu:~/core102_lab4$ sudo systemctl reload ssh  
dyahayaaddra11@ubuntu:~/core102_lab4$  
To see these additional updates r  
un: apt list --upgradable  
Enable ESM Apps to receive additi  
onal future security updates.  
See https://ubuntu.com/esm or run  
: sudo pro status  
*** System restart required ***  
Last login: Tue Sep 29 19:49:25 2  
026 from 192.168.113.128  
dyahayaaddra11@ubuntu:~$ sudo ssh  
d -T | grep -E 'permitrootlogin|p  
ubkeyauthentication|passwordauthe  
ntication|kbdinteractiveauthentic  
ation'  
[sudo: authenticate] Password: **  
permitrootlogin no  
pubkeyauthentication yes  
passwordauthentication yes  
kbdinteractiveauthentication no  
dyahayaaddra11@ubuntu:~$ sudo gre  
p -RniE '^[[:space:]]*(PasswordAu  
thentication|PubkeyAuthentication|Per  
mitRootLogin)' /etc/ssh/sshd_conf  
ig /etc/ssh/sshd_config.d/
```

RESPONSE:

Task 7: SSH Hardening and Recovery

In this task, I made the Ubuntu server's SSH login more secure. I stopped people from logging in directly as **root** and disabled **password-based login**, so users must use the approved SSH key instead.

Before making the changes, I backed up the SSH configuration and kept an existing login session open in case something went wrong. I also checked the configuration before restarting the SSH service.

During the process, I discovered another configuration file that was still allowing password login. I corrected it, tested the configuration again, and confirmed that SSH was now using the required security settings.

TASK 8 REPORT

EVIDENCE:

```
dyahayaaddra11@ubuntu:~$ sudo ss
-tlnp | grep ':8080' || echo "Temporary web service stopped"
[sudo: authenticate] Password: **

Temporary web service stopped
dyahayaaddra11@ubuntu:~$ sudo ufw
status numbered
Status: active

      To
Action  From      A
-----
[ 1] 22/tcp          A
ALLOW IN 192.168.113.0/24
      # CORE102 SSH
[ 2] 8080/tcp         A
ALLOW IN 192.168.113.128
      # CORE102 test web
[ 3] 23/tcp          D
DENY IN  Anywhere
      # Block Telnet
[ 4] Anywhere on vxlan.calico A
ALLOW IN Anywhere

[ 5] Anywhere          A
ALLOW OUT Anywhere on vxlan.calico
      (out)
[ 6] Anywhere on cali+  A
ALLOW IN  Anywhere
```

```

allow from 192.168.113.128 to any port 8080 proto tcp comment 'CORE102 t
est web'
Proceed with operation (y/n)? y
Rule deleted

```

```

dyahayaaddra11@ubuntu:~$ sudo ufw status numbered
Status: active

```

To	Action	From	
[1] 22/tcp CORE102 SSH	ALLOW IN	192.168.113.0/24	#
[2] 23/tcp Block Telnet	DENY IN	Anywhere	#
[3] Anywhere on vxlan.calico	ALLOW IN	Anywhere	
[4] Anywhere (out)	ALLOW OUT	Anywhere on vxlan.calico	(o
[5] Anywhere on cali+	ALLOW IN	Anywhere	
[6] Anywhere (out)	ALLOW OUT	Anywhere on cali+	(o
[7] 23/tcp (v6) Block Telnet	DENY IN	Anywhere (v6)	#
[8] Anywhere (v6) on vxlan.calico	ALLOW IN	Anywhere (v6)	
[9] Anywhere (v6) (out)	ALLOW OUT	Anywhere (v6) on vxlan.calico	
[10] Anywhere (v6) on cali+	ALLOW IN	Anywhere (v6)	
[11] Anywhere (v6) (out)	ALLOW OUT	Anywhere (v6) on cali+	(o

```

dyahayaaddra11@ubuntu:~$ sudo ss -tulpn
Netid State  Recv-Q Send-Q           Local Address:Port  Peer Address:Port
Process

```

```

users:(("mosquitto",pid=2126,fd=5))
tcp LISTEN 0 4096 0.0.0.0:22 0.0.0.0:*
users:(("sshd",pid=5772,fd=3),("systemd",pid=1,fd=295))
tcp LISTEN 0 4096 127.0.0.53:53 0.0.0.0:*
users:(("systemd-resolve",pid=623,fd=17))
tcp LISTEN 0 4096 127.0.0.1:19001 0.0.0.0:*
users:(("k8s-dqlite",pid=1658,fd=17))
tcp LISTEN 0 4096 127.0.0.1:9883 0.0.0.0:*
users:(("mosquitto",pid=2126,fd=7))
tcp LISTEN 0 4096 127.0.0.54:53 0.0.0.0:*
users:(("systemd-resolve",pid=623,fd=19))
tcp LISTEN 0 4096 *:9090 *:~
users:(("prometheus",pid=2133,fd=8))
tcp LISTEN 0 100 [::]:1883 [::]:~
users:(("mosquitto",pid=2126,fd=6))
tcp LISTEN 0 4096 *:16443 *:~
users:(("kubelite",pid=706329,fd=4))
tcp LISTEN 0 4096 *:10259 *:~
users:(("kubelite",pid=706329,fd=83))
tcp LISTEN 0 4096 *:10257 *:~
users:(("kubelite",pid=706329,fd=78))
tcp LISTEN 0 4096 [::]:22 [::]:~
users:(("sshd",pid=5772,fd=4),("systemd",pid=1,fd=302))
tcp LISTEN 0 4096 *:10250 *:~
users:(("kubelite",pid=706329,fd=96))
tcp LISTEN 0 511 *:80 *:~
users:(("httpd",pid=8572,fd=4),("httpd",pid=8571,fd=4),("httpd",pid=8570,fd=4),("httpd",pid=8554,fd=4))
tcp LISTEN 0 4096 *:25000 *:~
users:(("cluster-agent",pid=399632,fd=3))
dyahayaaddra11@ubuntu:~$

```

```

Session Actions Edit View Help
secure developer environments that ar
https://ubuntu.com/workshop
Expanded Security Maintenance for Applic
8 updates can be applied immediately.
To see these additional updates run: apt
Enable ESM Apps to receive additional fu
See https://ubuntu.com/esm or run: sudo

** System restart required **

Last login: Tue Sep 29 16:22:33 2026 from
dyahayaaddra11@ubuntu:~$ ls
core102_lab2 core102_lab4
dyahayaaddra11@ubuntu:~$ cd core102_lab4
dyahayaaddra11@ubuntu:~/core102_lab4$ ls
web
dyahayaaddra11@ubuntu:~/core102_lab4$ su
tc/ssh/sshd_config.core102.bak
[sudo: authenticate] Password:
dyahayaaddra11@ubuntu:~/core102_lab4$ su
d/90-core102-hardening.conf
dyahayaaddra11@ubuntu:~/core102_lab4$ su
dyahayaaddra11@ubuntu:~/core102_lab4$ su
dyahayaaddra11@ubuntu:~/core102_lab4$

users:(("mosquitto",pid=2126,fd=5)
tcp LISTEN 0 4096
users:(("sshd",pid=5772,fd=3),("sy
tcp LISTEN 0 4096
users:(("systemd-resolve",pid=623,
tcp LISTEN 0 4096
users:(("k8s-dqlite",pid=1658,fd=1
tcp LISTEN 0 4096
users:(("mosquitto",pid=2126,fd=7)
tcp LISTEN 0 4096
users:(("systemd-resolve",pid=623,
tcp LISTEN 0 4096
users:(("prometheus",pid=2133,fd=8
tcp LISTEN 0 100
users:(("mosquitto",pid=2126,fd=6)
tcp LISTEN 0 4096
users:(("kubelite",pid=706329,fd=4
tcp LISTEN 0 4096
users:(("kubelite",pid=706329,fd=8
tcp LISTEN 0 4096
users:(("kubelite",pid=706329,fd=7
tcp LISTEN 0 4096
users:(("sshd",pid=5772,fd=4),("sy
tcp LISTEN 0 4096
users:(("kubelite",pid=706329,fd=9
tcp LISTEN 0 511
users:(("httpd",pid=8572,fd=4),("h
,fd=4),("httpd",pid=8554,fd=4))
tcp LISTEN 0 4096
users:(("cluster-agent",pid=399632
dyahayaaddra11@ubuntu:~$

Swap usage: 0%
* Canonical Workshop gives develop
ers fast, composable, reproducible,
and
secure developer environments th
at are perfect for agentic workflow
s.
https://ubuntu.com/workshop
Expanded Security Maintenance for A
pplications is not enabled.
8 updates can be applied immediatel
y.
To see these additional updates run
: apt list --upgradable
Enable ESM Apps to receive addition
al future security updates.
See https://ubuntu.com/esm or run:
sudo pro status
** System restart required **

Last login: Tue Sep 29 19:59:34 202
6 From 192.168.113.128
dyahayaaddra11@ubuntu:~$

passwordauthentication
kdbinteractiveauthenti
dyahayaaddra11@ubuntu:
/etc/ssh/sshd_config.d
nit.conf /etc/ssh/50-c
onf.core102.bak
dyahayaaddra11@ubuntu:
-i 's/'[:space:]]*Pa
ntication[[:space:]]\+
dAuthentication no/' /
d_config.d/50-cloud-in
dyahayaaddra11@ubuntu:
d -t
dyahayaaddra11@ubuntu:
temctl reload ssh
dyahayaaddra11@ubuntu:
d -T | grep -E 'permit
ubkeyauthentication|pa
ntication|kdbinteracti
ation'
permitrootlogin no
pubkeyauthentication y
passwordauthentication
kdbinteractiveauthenti
dyahayaaddra11@ubuntu:
dyahayaaddra11@ubuntu:
dyahayaaddra11@ubuntu:
dyahayaaddra11@ubuntu:
dyahayaaddra11@ubuntu:
dyahayaaddra11@ubuntu:
dyahayaaddra11@ubuntu:
~$

```

RESPONSE:

Task 8: Cleanup and Final Review

In this task, I cleaned up the temporary changes made during the lab. I stopped the temporary web server running on port **8080** and removed its firewall rule.

I checked the ssh on port 22 and it working and the telnet on port 23 is blocked by the firewall, and I also checked the connectivity it is reachable.

Finally, I tested SSH again from Kali using the Ed25519 key and successfully logged into the Ubuntu server.

In simple terms: I removed the temporary things I created for testing, checked that the firewall was still protecting the server, and confirmed that I could still securely access the server.